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1. Write a C program using nested loops to print the following pattern:

   1
   1 1
   1 1 1
   1 1 1 0
   1 1 1 0 1
   1 1 1 0 1 1
   1 1 1 0 1 1 1 1 1 1 0 1 1 1 0

looppattern.c | patternlooppss.c | su

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```c
#include <stdio.h>
int main(){
    for(int i=1;i<=8;i++){
        for(int j=1;j<=i;j++){
            if(j==4||j==8)
                printf("0");
            else
                printf("1");
        }
        printf("\n");
    }
}
```

```
1
11
111
1110
11101
111011
1110111
11101110
```

2. Write a C program using nested loops to print the following pattern:

   1
   1 0
   1 0 1
   1 0 1 0
   1 0 1 0 1

looppattern.c | patternlooppss.c

folderofc.c > patternlooppss.c

```c
#include <stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=i;j++){
            if(j==2||j==4)
                printf("0 ");
            else
                printf("1 ");
        }
        printf("\n");
    }
}
```

```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
PS C:\Users\I C\VS\folderofc.c>
```

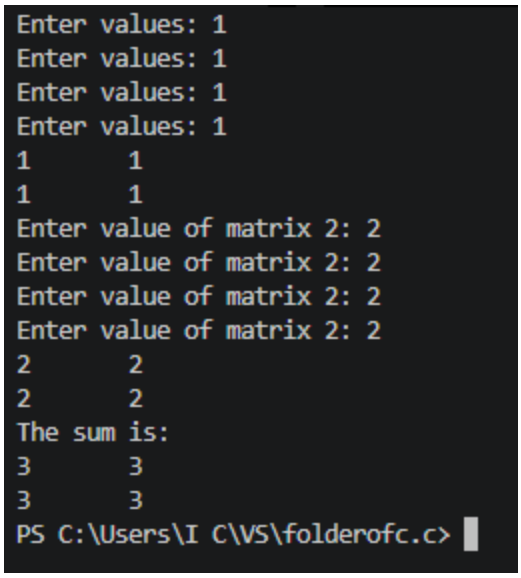
3. Write a C program to input two 3×3 matrices using separate 2D arrays, add their corresponding elements using nested loops, store the results in a third matrix, and display both input matrices and the resulting sum matrix.

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```c
#include<stdio.h>
int main(){
    int arr[2][2],arr1[2][2],sum[2][2];

    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            printf("Enter values: ");
            scanf("%d",&arr[i][j]);
        }
    }
    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            printf("%d\t",arr[i][j]);
        }
        printf("\n");
    }
    //second matrix ka input and output
      for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            printf("Enter value of matrix 2: ");
            scanf("%d",&arr1[i][j]);
        }
    }
    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            printf("%d\t",arr1[i][j]);
        }
        printf("\n");
    }
    //sum ke liye
    printf("The sum is: \n");
    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            sum[i][j]=arr[i][j]+arr1[i][j];
            printf("%d\t",sum[i][j]);
        }
        printf("\n");
    }
}
```



4. Write a C program using a 2D array to store the Mid and Final exam marks of 4 students (each row is a student, with one column for Mid and one for Final). Input the marks using nested loops, then calculate and display: 1. The total marks of each student (Mid + Final) 2. The highest marks in Mid, the highest marks in Final, and the highest marks overall across both 3. The lowest marks in Mid, the lowest marks in Final, and the lowest marks overall across both 4. The average of Mid marks and the average of Final marks, calculated separately

looppattern.c | patternlooppss.c | sumofmatrix2.c | arraymarks.c

folderofc.c > arraymarks.c

```c
#include <stdio.h>
int main() {
    int marks[4][2];
    int midTotal = 0;
    int finalTotal = 0;
    int highestMid, highestFinal;
    int lowestMid, lowestFinal;

    for(int i=0;i<4;i++){
        printf("Enter Mid marks for student %d:", i + 1);
        scanf("%d",&marks[i][0]);

        printf("Enter Final marks for student %d: ", i + 1);
        scanf("%d",&marks[i][1]);
    }

    highestMid = marks[0][0];
    lowestMid = marks[0][0];

    highestFinal = marks[0][1];
    lowestFinal = marks[0][1];

    for(int i = 0; i < 4; i++){

        int total=marks[i][0]+marks[i][1];

        printf("Student %d total = %d\n", i + 1, total);

        midTotal = midTotal + marks[i][0];
        finalTotal = finalTotal + marks[i][1];

        if(marks[i][0] > highestMid)
            highestMid = marks[i][0];
        if(marks[i][1] > highestFinal)
            highestFinal = marks[i][1];
        if(marks[i][0] < lowestMid)
            lowestMid = marks[i][0];
        if(marks[i][1] < lowestFinal)
            lowestFinal = marks[i][1];
```

```
        if(marks[i][0] < lowestMid)
            lowestMid = marks[i][0];
        if(marks[i][1] < lowestFinal)
            lowestFinal = marks[i][1];
    }

    printf("\nHighest Mid = %d", highestMid);
    printf("\nHighest Final = %d", highestFinal);

    if(highestMid > highestFinal)
        printf("\nHighest Overall = %d", highestMid);
    else
        printf("\nHighest Overall = %d", highestFinal);

    printf("\nLowest Mid = %d", lowestMid);
    printf("\nLowest Final = %d", lowestFinal);

    if(lowestMid < lowestFinal)
        printf("\nLowest Overall = %d", lowestMid);
    else
        printf("\nLowest Overall = %d", lowestFinal);

    printf("\nAverage Mid = %.2f", (float)midTotal / 4);
    printf("\nAverage Final = %.2f", (float)finalTotal / 4);
}
```

```
Enter Mid marks for student 1:10
Enter Final marks for student 1: 15
Enter Mid marks for student 2:20
Enter Final marks for student 2: 25
Enter Mid marks for student 3:12
Enter Final marks for student 3: 32
Enter Mid marks for student 4:14
Enter Final marks for student 4: 21
Student 1 total = 25
Student 2 total = 45
Student 3 total = 44
Student 4 total = 35

Highest Mid = 20
Highest Final = 32
Highest Overall = 32
Lowest Mid = 10
Lowest Final = 15
Lowest Overall = 10
Average Mid = 14.00
Average Final = 23.25
PS C:\Users\I C\VS\folderofc.c>
```

5. A university stores the marks of students in a 3D array int marks[2][3][3], where the first index is the semester (2 semesters), the second index is the student (3 students) and the third index is the subject (3 subjects). On paper, draw the array as 2 layers (one per semester) and label every cell with its index marks[semester][student][subject]. Decide what to insert: fill every cell with a realistic mark of your choice (0-100), writing the name of the semester, each student and each subject on the margins of your diagram. Then, from your diagram, write down: (a) the index and value of the highest mark, (b) the marks of student 1 in semester 2 across all subjects, and (c) the total marks of each student in semester 1.

Q5)

int marks [2][3][3].

Semester 1	(marks[0])		
	Matt	Prog	eng.
student 1	75	82	68
student 2	90	77	85
student 3	64	88	72

Semester 2	(Marks [1])		
	Maths	Prog	Eng
student 1	92	86	79
student 2	81	75	88
student 3	73	84	91

a) highest marks:-

95 → marks [1][1][1]

b) student 1 in sem 2 :-

marx [1][0][0] = 92

" [1][0][1] = 86

" [1][0][2] = 72.

c) Total mark of each student in sem 1 :-

Student 1 : 75+82+68 = 225

" 2 : 90+77+85 = 252

" 3 : 64+88+72 = 224.

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6. On paper, draw the 3D array int arr[2][3][3] as 2 layers (pages), each layer containing a 3×3 grid. Label every cell with its index in the form arr[i][j][k] and label the layers, rows and columns. Fill the cells with any values of your choice.

Q6)

int arr[2][3][3];

arr[0]

	C0	C1	C2
R0	10	20	30
R1	40	50	60
R2	70	80	90

arr[1]

	C0	C1	C2
R0	100	110	120
R1	130	140	150
R2	160	170	180

Index & values:

arr[0][0][0] = 10 arr[0][0][1] = 20

arr[0][1][0] = 40 arr[0][1][1] = 50

arr[0][2][0] = 70 arr[0][2][1] = 80

arr[0][0][2] = 30

arr[0][1][2] = 60

arr[0][2][2] = 90.

arr[1][0][0] = 100 arr[1][0][1] = 110

arr[1][1][0] = 130 arr[1][1][1] = 140

arr[1][2][0] = 160 arr[1][2][1] = 170.

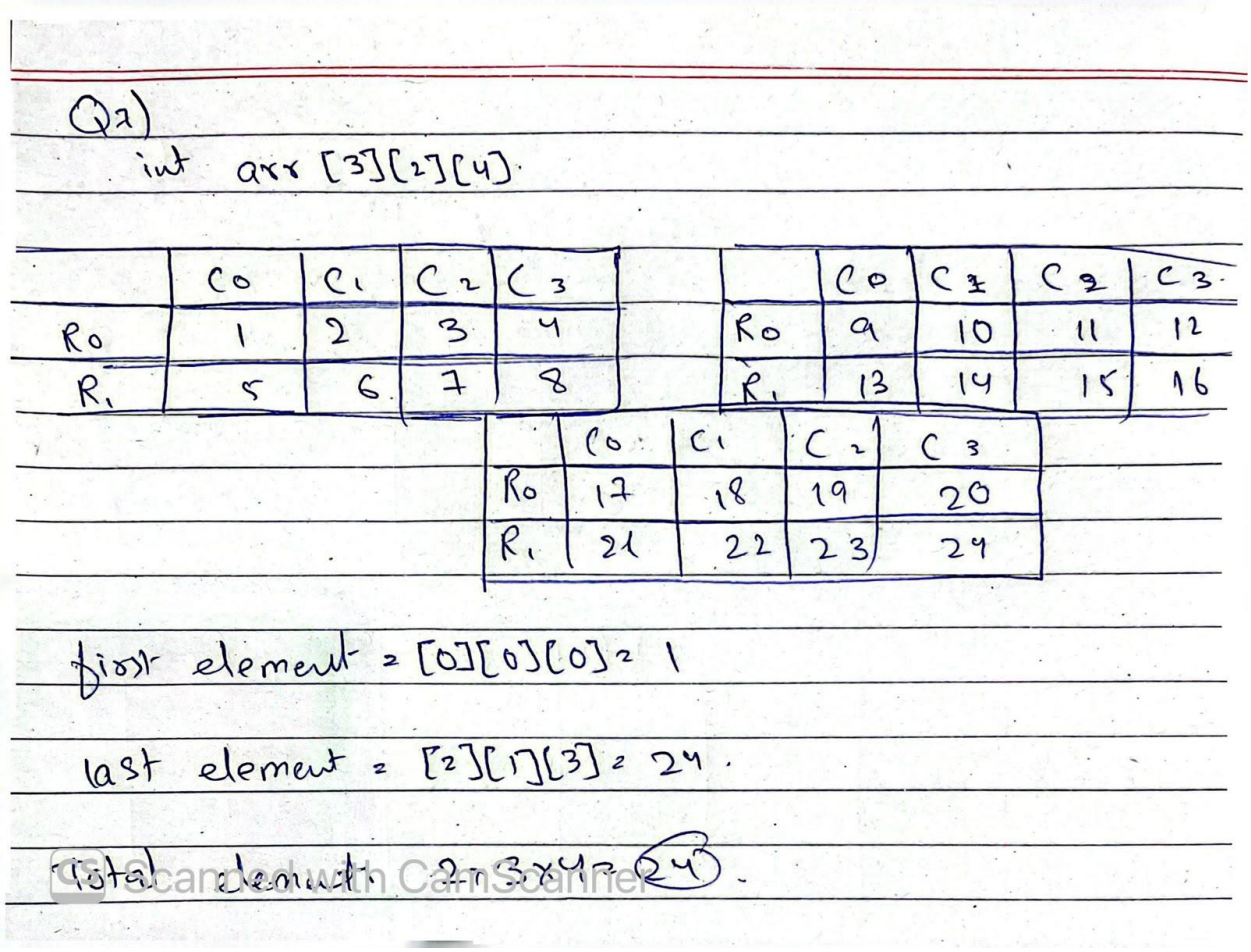
arr[1][0][2] = 120.

arr[1][1][2] = 150

arr[1][2][2] = 180.

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7. On paper, draw the 3D array int arr[3][2][4] as 3 layers (pages), each layer containing 2 rows and 4 columns. Show the index of every cell, fill the array with values, and then write the index of the first and last element of each layer and the total number of elements in the array.



Q7)

int arr [3][2][4].

	C0	C1	C2	C3
R0	1	2	3	4
R1	5	6	7	8

	C0	C1	C2	C3
R0	9	10	11	12
R1	13	14	15	16

	C0	C1	C2	C3
R0	17	18	19	20
R1	21	22	23	24

first element = [0][0][0] = 1

last element = [2][1][3] = 24.

Total elements = 2 × 3 × 4 = 24.

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